

Chapter 6: Anatomy of Flowering Plants

Revision Notes — High-yield summary for quick recall before exams

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8 Topics

The Tissue System

- 3 tissue systems: Epidermal, Ground, Vascular
- Stomatal apparatus = guard cells + subsidiary cells
- Trichomes (stem) vs Root hairs (root)
- Open (cambium) vs Closed vascular bundle
- Conjoint (stem/leaf) vs Radial (root)

Dicot Root

- Epiblema = outer layer with root hairs
- Cortex → Endodermis (Casparian strips) → Stele
- Radial vascular bundle, exarch, 2–4 xylem patches
- Pith small & inconspicuous
- Pericycle initiates lateral roots & cambium

Monocot Root

- Same 4 regions as dicot root
- Vascular bundle radial, exarch, polyarch (>6 patches)
- Pith large & well developed
- No cambium → no secondary growth ever

Dicot Stem

- Hypodermis = collenchyma (mechanical strength)
- Ground tissue = cortex + pith, divided by ring of VBs
- Vascular bundles in a ring: conjoint, open, endarch
- Pericycle = sclerenchyma patches
- Medullary rays between vascular bundles

Monocot Stem

- Hypodermis = sclerenchyma
- Ground tissue undifferentiated (no distinct cortex/pith)
- Vascular bundles scattered, conjoint, closed, endarch
- Bundle sheath of sclerenchyma around each VB

- Phloem parenchyma absent; pith absent

Dicot Leaf

- Dorsiventral (bifacial) leaf
- Adaxial (upper) & abaxial (lower) epidermis, both with cuticle
- More stomata on abaxial surface
- Mesophyll differentiated: palisade + spongy parenchyma
- Vascular bundle conjoint & closed, with bundle sheath

Monocot Leaf

- Isobilateral leaf
- Stomata roughly equal on both surfaces
- Mesophyll NOT differentiated into palisade/spongy
- Bulliform cells on adaxial epidermis control rolling
- Vascular bundle conjoint & closed, endarch

Dicot vs Monocot

- Root: mono/oligoarch small pith (dicot) vs polyarch large pith (monocot)
- Stem: ring VBs + pith (dicot) vs scattered VBs, no pith (monocot)
- Leaf: dorsiventral, differentiated mesophyll (dicot) vs isobilateral, undifferentiated (monocot)
- Golden rule: Open+Radial = dicot root; Open+Conjoint = dicot stem; Closed+Conjoint = monocot stem/leaf